

FALL 2026



EE3003/EE3010 Analog and Digital Communications

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Department of Electrical Engineering

FALL 2026

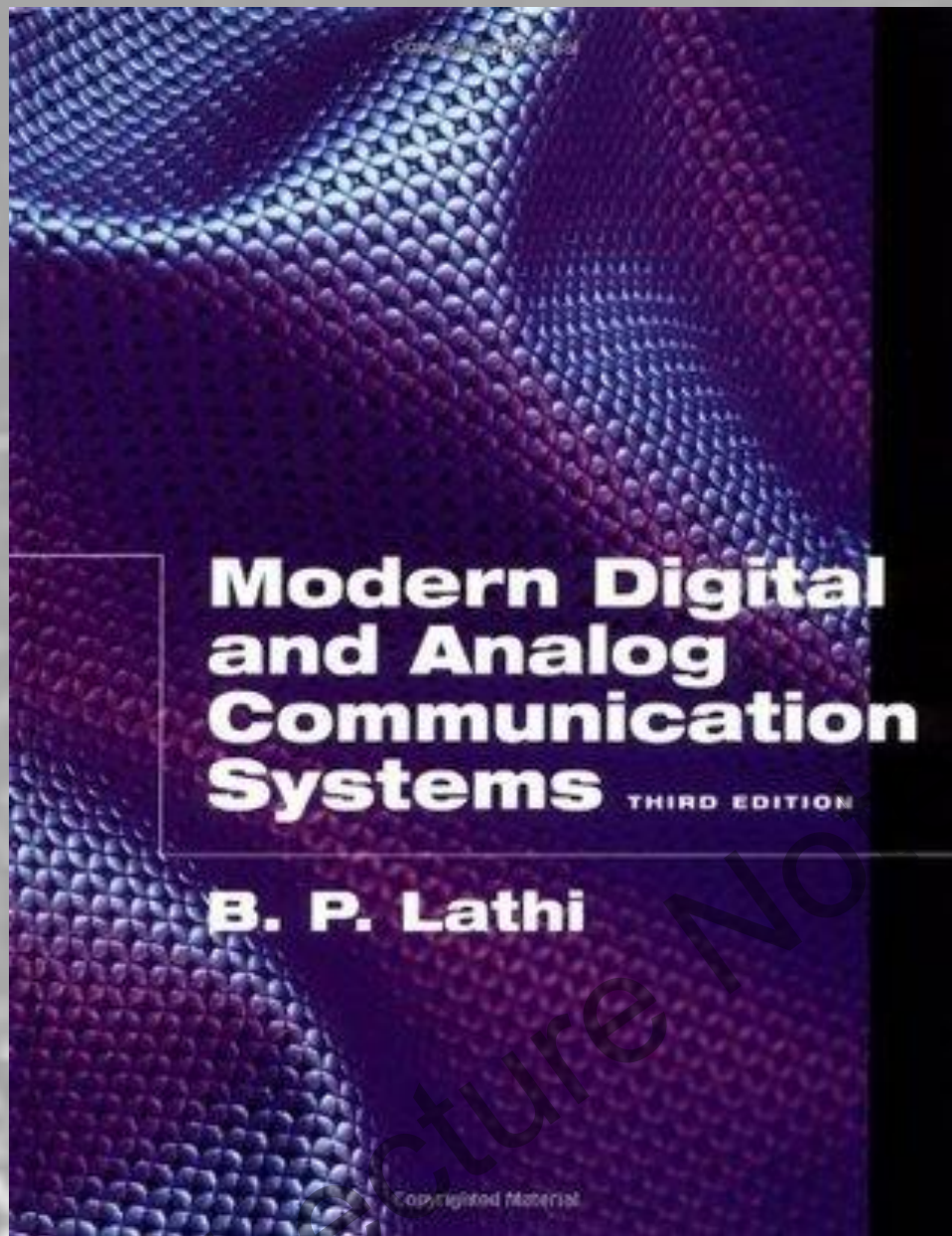
Course Objectives

- To provide students with a comprehensive understanding of both analog and digital communication systems
- To equip students with the theoretical knowledge and practical engineering skills required to design, calculate, and evaluate transmission techniques
- To study analog modulation schemes (AM, FM), baseband conversion methods (sampling, PCM)
- To study digital modulation systems (ASK, PSK, FSK), and the foundational limits of channel capacity under noise conditions.

FALL 2026

Course Learning Outcomes

No.	Course Learning Outcome (CLO) Statements	Assessment Tools
1	Explain the fundamental models of communication systems, signal transmission characteristics, distortions, and energy/power spectral densities.	Q1, A1, M1
2	Calculate parameters like bandwidth, power efficiency, transmission efficiency, and modulation index for linear (AM, SSB, VSB, QAM) and non-linear (FM, PM) analog modulation systems.	Q2, M1, M2
3	Apply the sampling theorem and analog-to-digital pulse conversion techniques (PAM, PWM, PPM, PCM, DPCM, DM) for baseband processing.	A2, M2, F
4	Analyze the performance, error probabilities, and spectral characteristics of digital communication systems while evaluating fundamental information theory trade-offs.	Q3, A3(PBL), F



Text Book

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Course Learning Objectives

Upon completion of EE3010 ADC, a student will be able to:

- Understand the differences between analog and digital communication systems.
- Understand signal multiplexing, modulation and demodulation
- Application of Sampling Theorem to ADC conversion and the limitations
- Gain an appreciation of wireless communication systems



Introduction to Communication System

A “**communication system**” is an apparatus that conveys information from a source (the transmitter) to a destination (the receiver) over a channel (the propagation medium carrying the signal).

A vertical section of the slide features a detailed, grayscale wood grain texture. The grain lines are wavy and concentric, creating a sense of depth and organic pattern. This texture occupies the left half of the slide, while the right half is white.

Common Sources of Information

- **Audio/voice** –information in acoustic form
- **Text messages** –written text sent in digital format
- **Data**–computer generated information in digital format
- **Video**--electronic representation of images or pictures

Signals Carry Content in Communication Systems

- **Data, messages, and information** (*i.e.*, useful content) are sent from transmitter to receiver over a channel often using **electrical signals**.
- A **signal** is a sequence of values, or symbols, encoding the transmitted message.

Today's communication systems mostly use **electrical and optical signals** that are **time-varying, electrical quantities** (*e.g.*, voltages, currents, and electromagnetic field quantities) where time variation encodes (*i.e.*, represents) data, messages, and information.

Signals Carry Content in Communication Systems

- Important non-electrical signals include acoustic (voice and music).
- A defined language or code is required between sender and receiver for communication.
 - For digital signals we use various digital codes (e.g., binary).

What makes a signal to be **information**?

- 1) It is accurate and timely,
- 2) Has a specific and organized purpose or focus, and
- 3) Results in increased understanding or decreased uncertainty.



The Four Great Enablers of the Communication Age

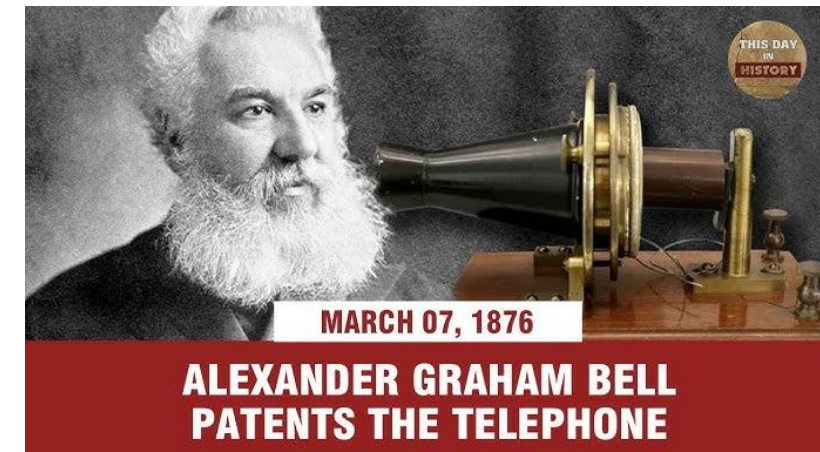
1. Harnessing of Electricity
2. Radio Waves
3. Digitization
4. Transistors and Integrated Circuits

1. Harnessing of Electricity

- In 1800, **Alessandro Volta** invented the voltaic pile, an early electric battery, which produced a steady electric current.
- In 1844, **Morse Telegraph** was successfully demonstrated.
- In 1876, **Graham Bell** patents the Telephone.

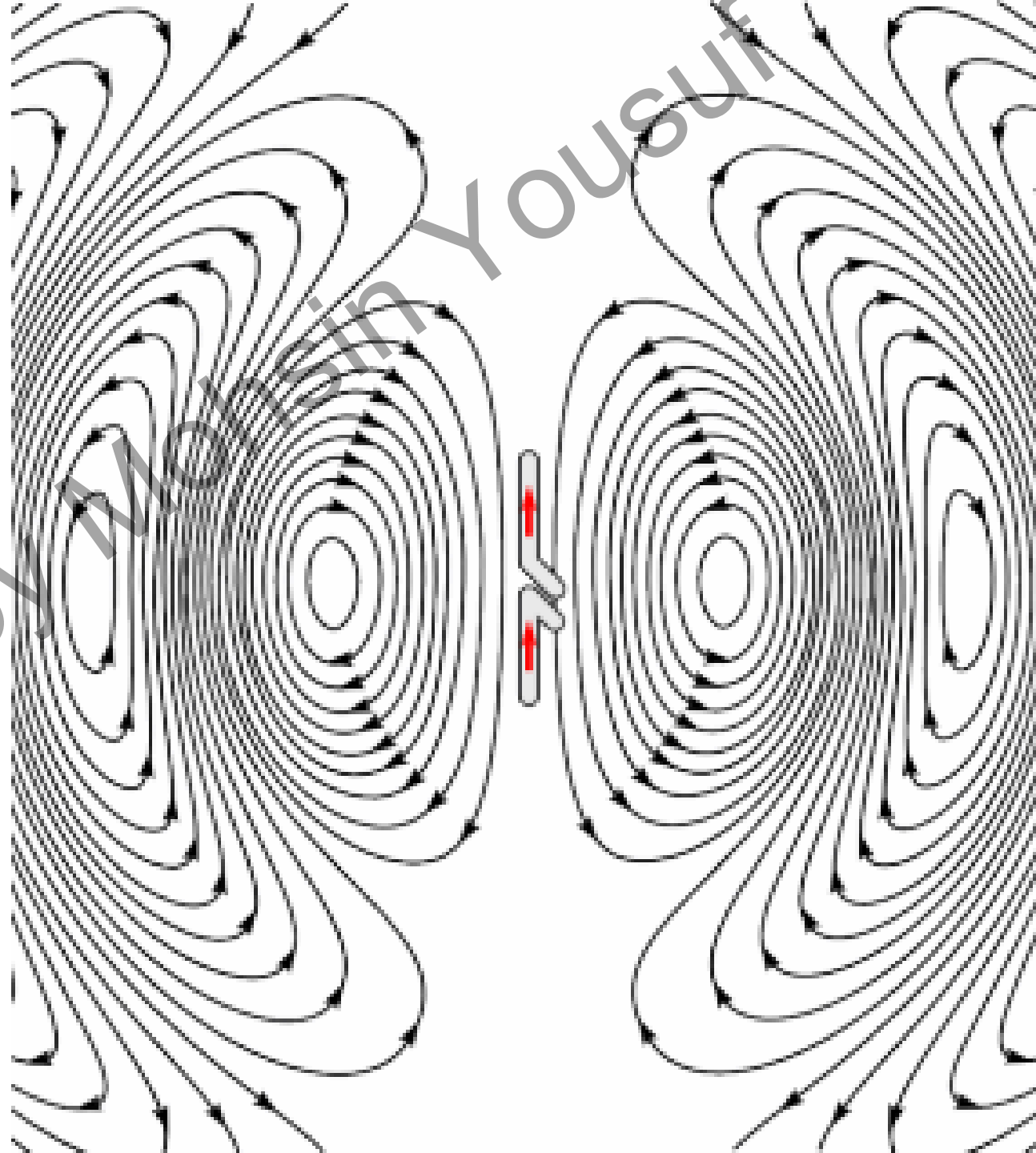


Volta battery at the [Tempio Voltiano](#) museum, Como



2. Radio Waves

- Began in 1896 with Wireless Telegraphy invented by Guglielmo Marconi



3. Digitization

- Started in 1940s with development of first digital computer, such as ENIAC
- American Scientist Claude Shannon's work on Information theory paved the way for digitizing information in late 1940s.
- Digitization accelerated in 1970s.



4. Transistors and ICs

Acronym	Name	Year	Transistor Count	Logic Gates
SSI	Small-Scale Integration	1964	1 to 10	1 to 12
MSI	Medium-Scale Integration	1968	10 to 500	13 to 99
LSI	Large-Scale Integration	1971	500 to 20,000	100 to 9999
VLSI	Very Large-Scale Integration	1980	20,000 to 1,000,000	10,000 to 99,999
ULSI	Ultra Large-Scale Integration	1984	1,000,000 to more	100,000 to more



Historical Review of Modern Telecommunications

1820

- First experiment of electric current causing magnetism (by Hans C. Oersted)

1831

- Discovery of induced current from electromagnetic radiation (by Michael Faraday)

1837

- Invention of Morse code by Samuel F. B. Morse

1864

- Theory of electromagnetic waves developed by James C. Maxwell

1876

- Invention of telephone by Alexander G. Bell

1887

- Detection of electromagnetic waves by Heinrich Hertz

Historical Review of Modern Telecommunications

1896

- Wireless telegraphy (radio telegraphy) patented by Guglielmo Marconi

1906

- First amplitude modulation radio broadcasting (by Reginald A. Fessenden)

1935

- First FM radio demonstration (by Edwin H. Armstrong)

1947

- Cellular concept first proposed at Bell Labs

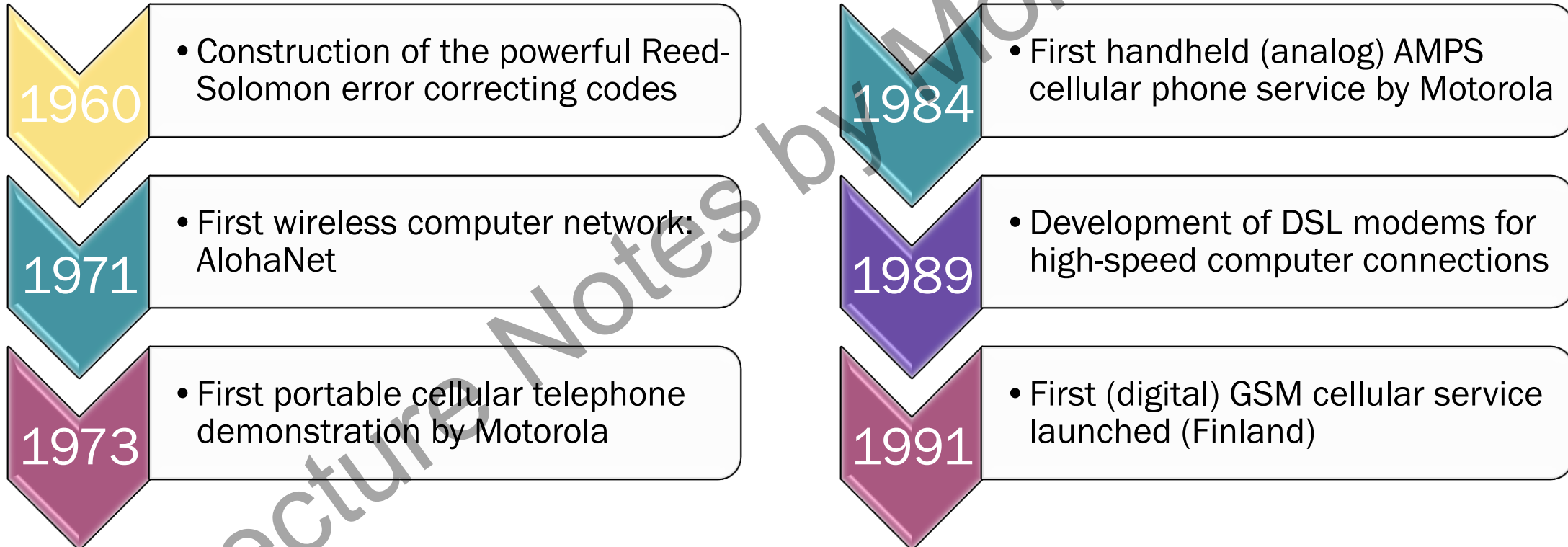
1948

- First major information theory paper published by Claude E. Shannon + Invention of Transistor

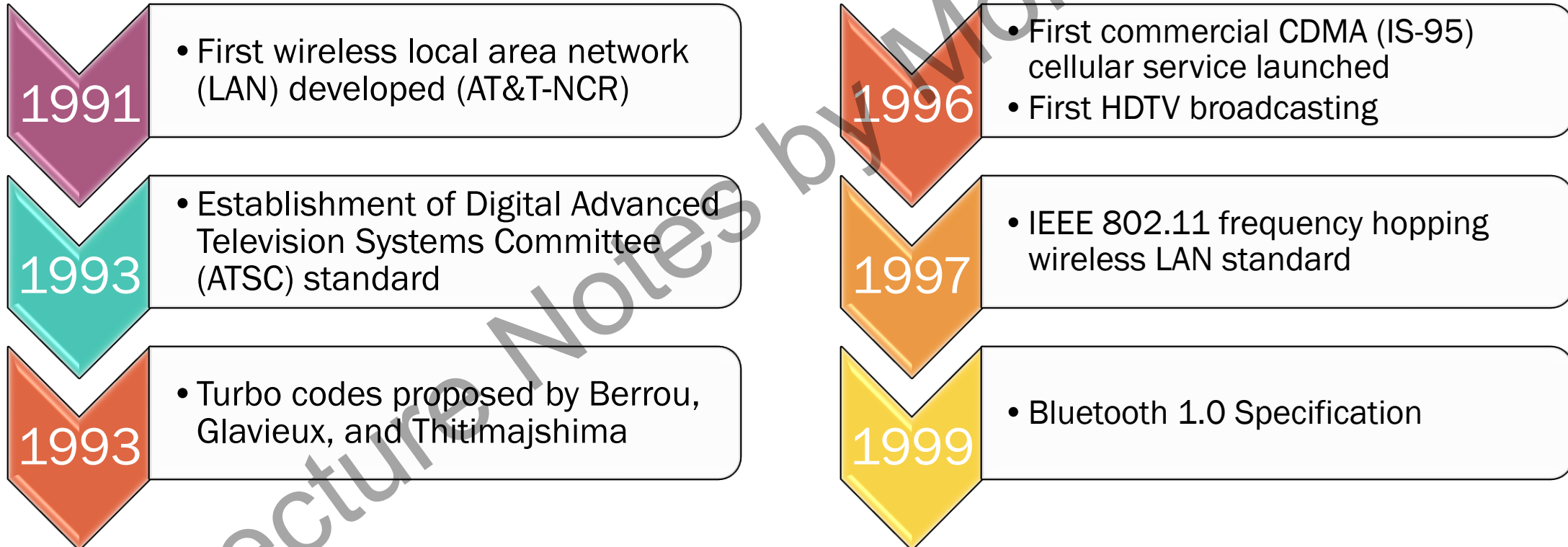
1958

- Integrated circuit proposed by Jack Kilby (Texas Instruments)

Historical Review of Modern Telecommunications



Historical Review of Modern Telecommunications



Historical Review of Modern Telecommunications

2000

- First 3G cellular service launched

2007

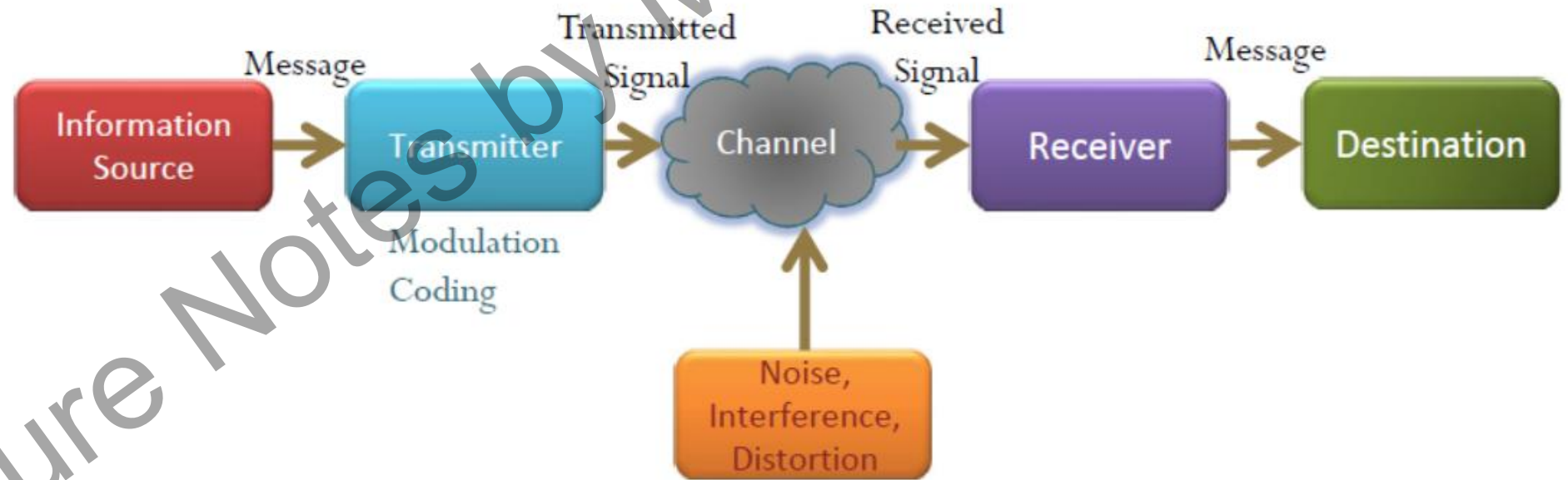
- First iPhone introduced by Apple CEO Steve Jobs

2008

- 4G-LTE cellular standard published by 3GPP in Release 8

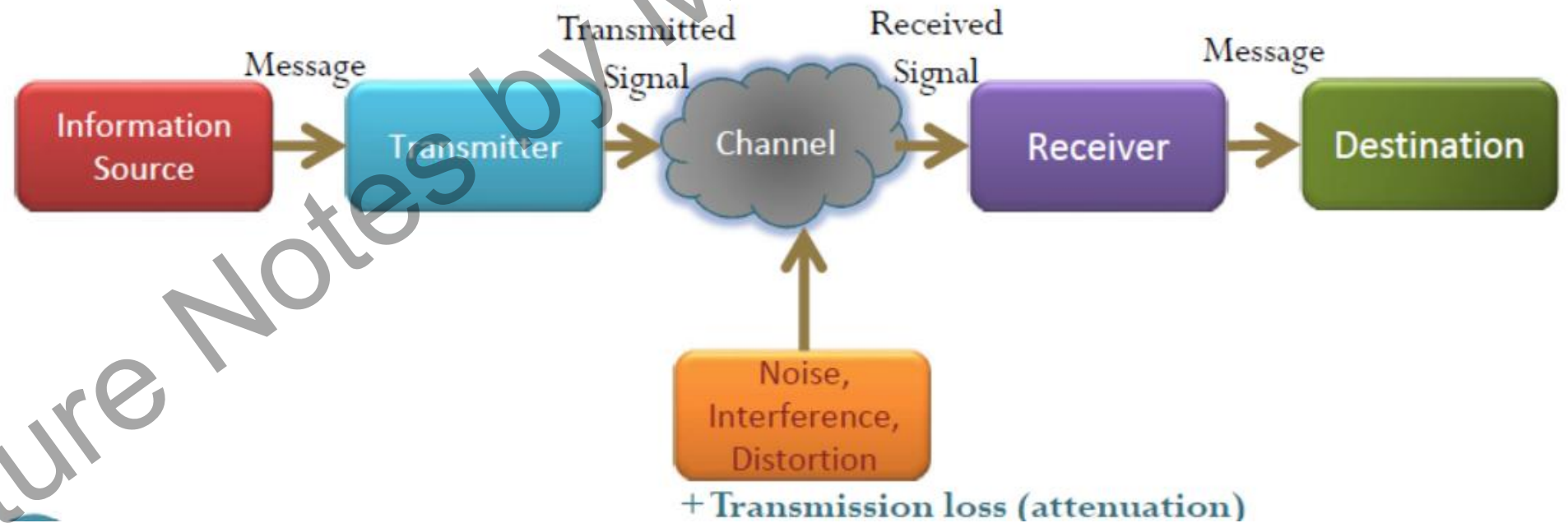
Basic Elements of Communication (1)

- **Information source**: produce a **message** { Analog (continuous)
Digital (discrete)
- **Transmitter**: operate on the message to create a **signal** (in an appropriate form) which can be sent through a channel



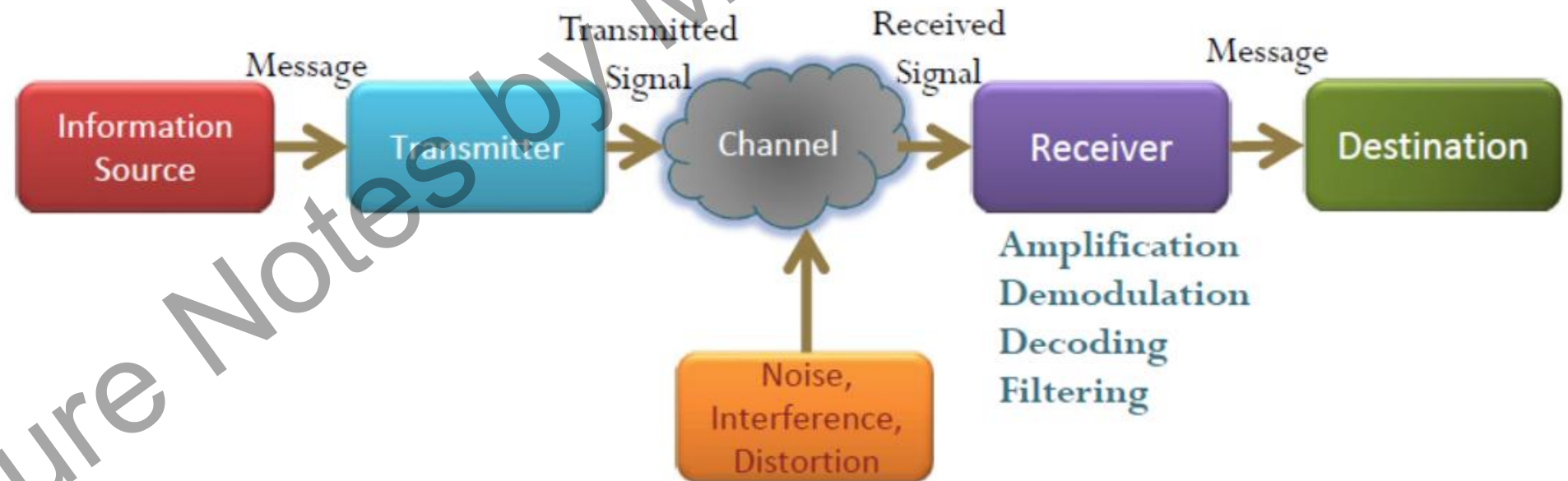
Basic Elements of Communication (2)

- **Channel**: the medium over which the signal, carrying the information that composes the message, is sent
- **Channel impairments**: Unwanted undesirable effects which corrupt the message

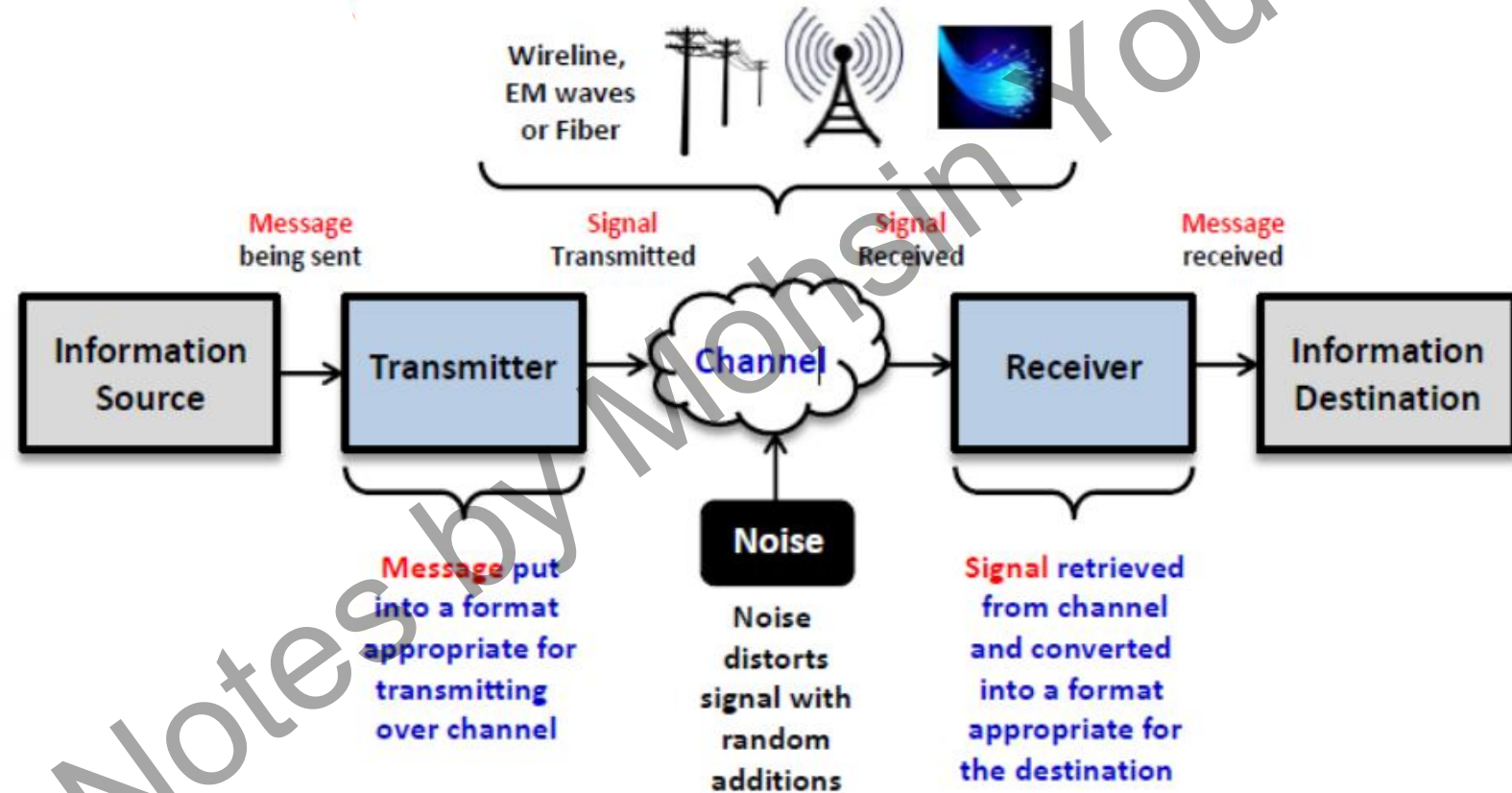


Basic Elements of Communication (3)

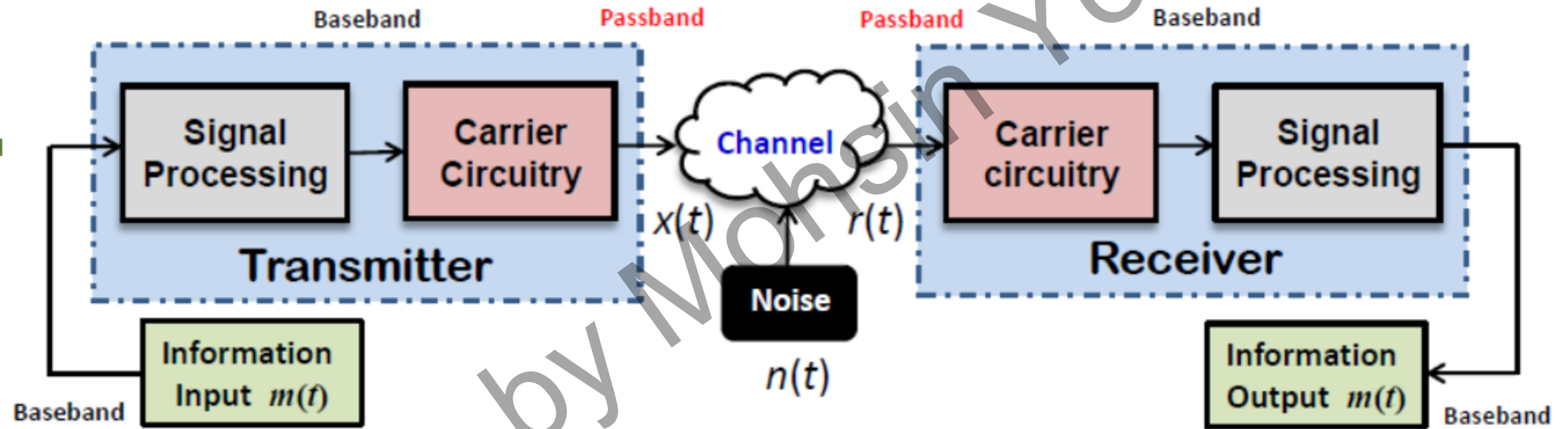
- **Receiver**: transform the received signal back into the message intended for delivery
- **Destination**: a person or a machine, for whom or which the message is intended



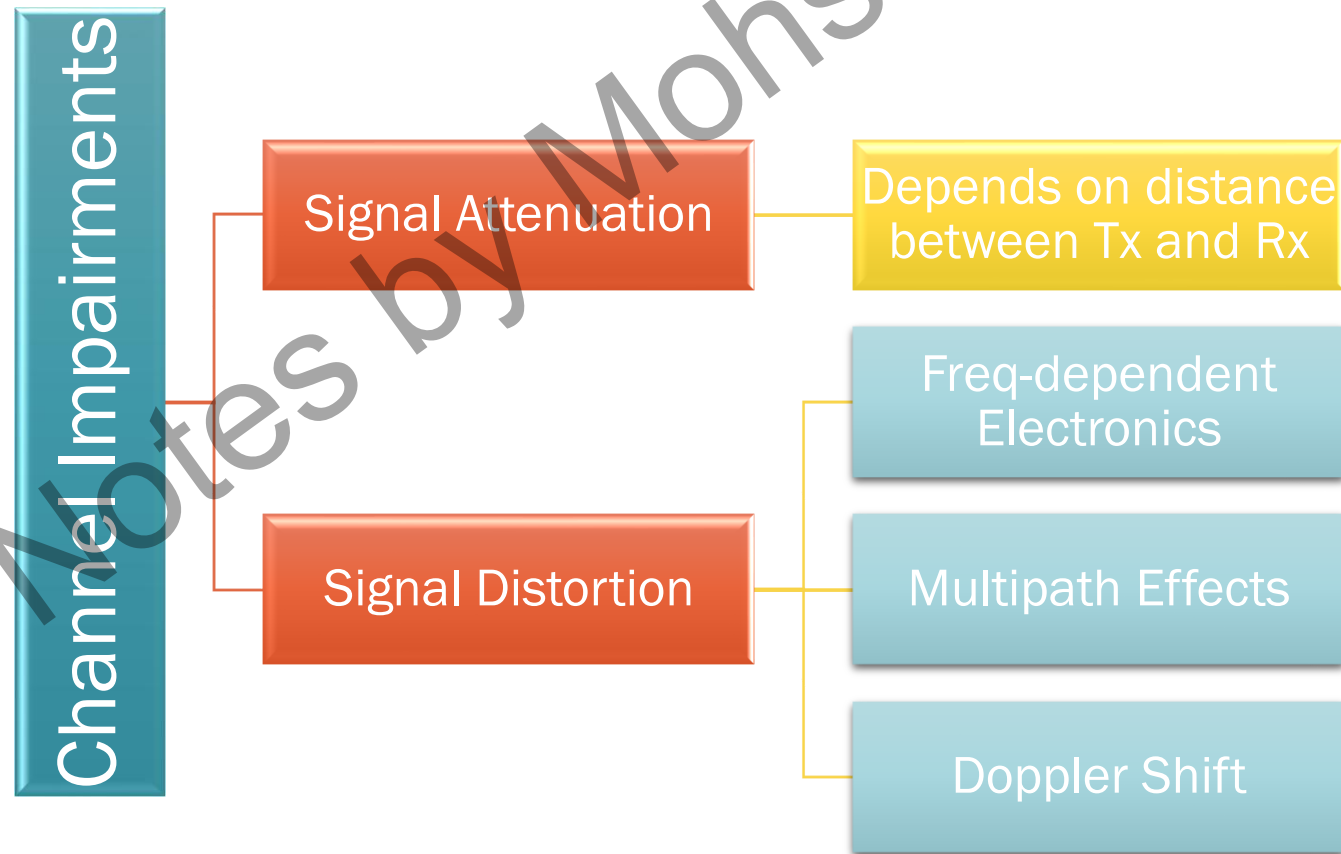
Shannon-Weaver Model for Communication



Communication System in Greater Detail



Channel Distortions and Noises: Design Challenge



Channel Distortions and Noises: Design Challenge

Channel Impairments

Signal Distortion

Linear Distortion

Non-linear Distortion

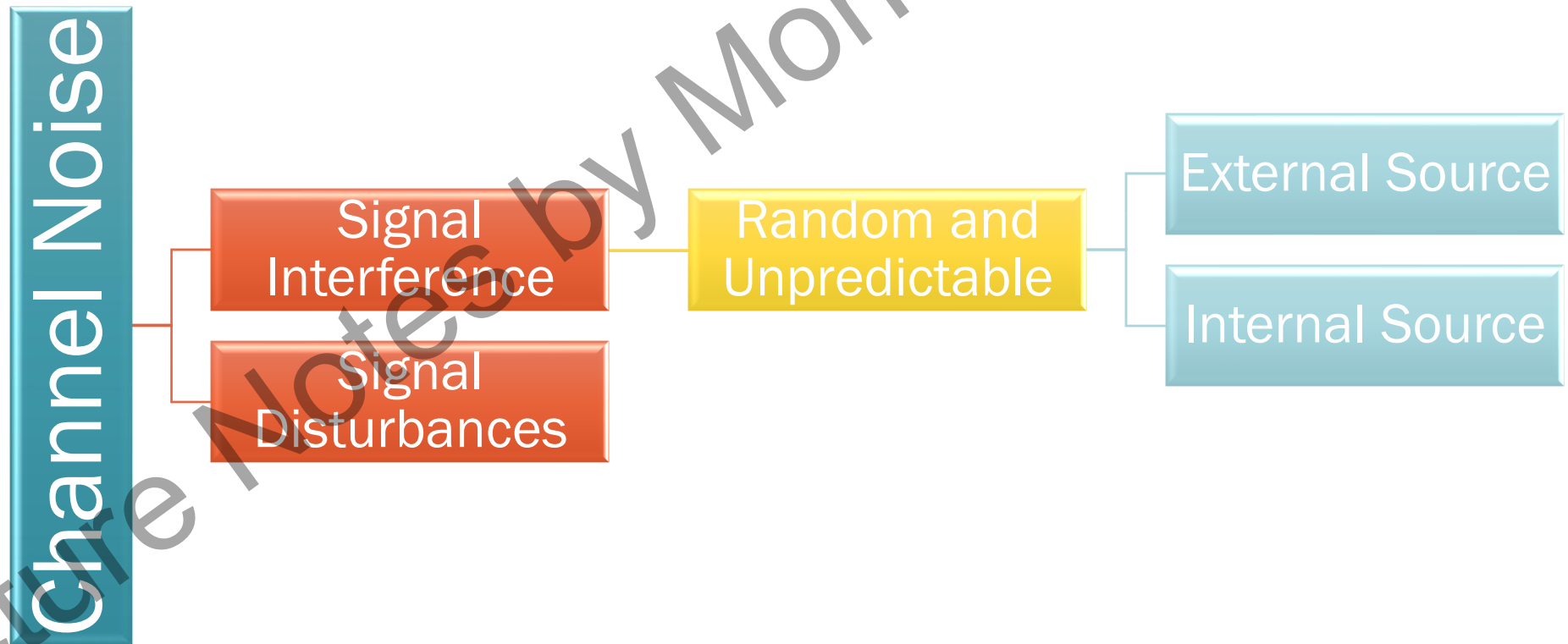
Frequency-Selective attenuation

Attenuation as a func of amplitude

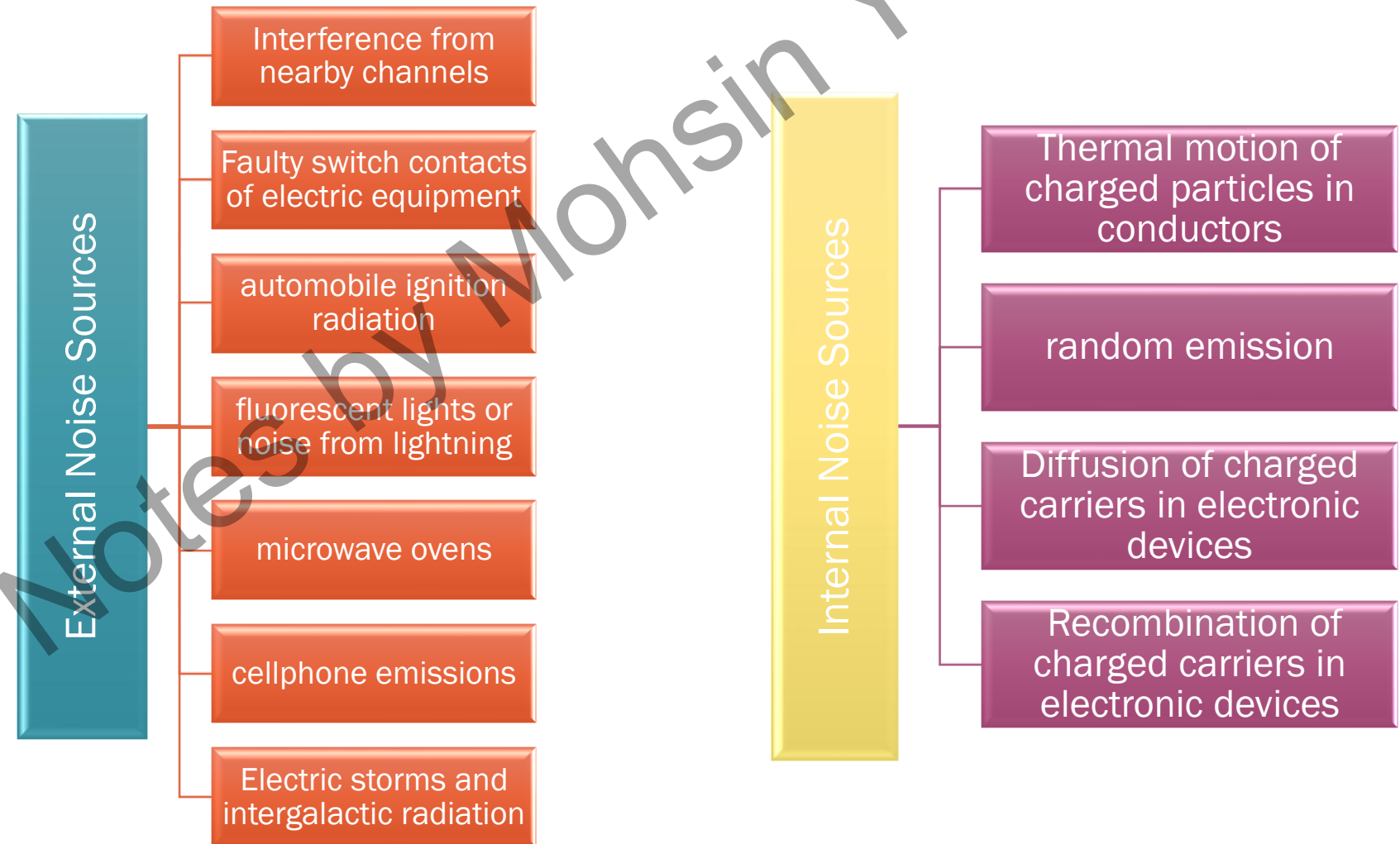
Equalizer : Solution to Signal Distortion

- A circuit with gain and phase characteristics complementary to the those of the channel

Channel Distortions and Noises: Design Challenge



Channel Distortions and Noises: Design Challenge



Communication System Parameters

SNR, Bandwidth, and rate of communication

The **fundamental parameters** that control the rate and quality of information transmission are:

1. Channel Bandwidth 'B'
 2. Signal Power 'S'
- The rate of information transmission is directly proportional to B

Bandwidth	When referred to signal	When referred to channel
	Difference between upper and lower frequency limits	Same
	Range of frequencies that the signal occupies	Range of frequencies that the medium can support
	Portion of the EM spectrum occupied by the signal	Max amount of information that the channel can carry

Communication System Parameters

SNR, Bandwidth, and rate of communication

“the rate of information transmission is directly proportional to B”, how?

- Consider the possibility of increasing the speed of information transmission by time compression of the signal.
- If a signal is *compressed in time* by a factor of 2, it can be transmitted in half the time, and the *speed of transmission is doubled*.
 - **Compression in time** corresponds to **expansion in frequency**.
- To transmit this compressed signal without distortion, the channel bandwidth ‘B’ must also be doubled.

Type of the Channel	Frequency Range (Approx.)
Twisted Pair	1 – 100 MHz
Coaxial Cable	0 – 750 MHz
Microwave	1 – 30 GHz
Satellite	1 – 40 GHz
Fiber Optics	180 – 330 THz

Communication System Parameters

SNR, Bandwidth, and rate of communication

“the signal-to-noise ratio” (SNR) is defined as the ratio of signal power to noise power.

- The signal power plays a dual role in information transmission.
 - I. It is related to the quality of transmission. Increasing S , i) reduces the effect of channel noise, ii) decreases uncertainty (reduces the probability of error) and iii) allows transmission over a longer distance.
 - II. The channel bandwidth B and the signal power S are exchangeable, i.e., a trade off exists between them.
- The relationship between B and SNR is exponential.

$$SNR_2 \cong SNR_1^{B_1/B_2}$$

- ❖ The above equation sets an upper bound, not all the systems are capable of achieving this bound.

Communication System

Power Limited Vs Band Limited

Power Limited	Band Limited
In space communication systems , infinite bandwidth is available but the power is limited.	A typical telephone channel has a limited bandwidth (3 kHz), but a lot of power is available

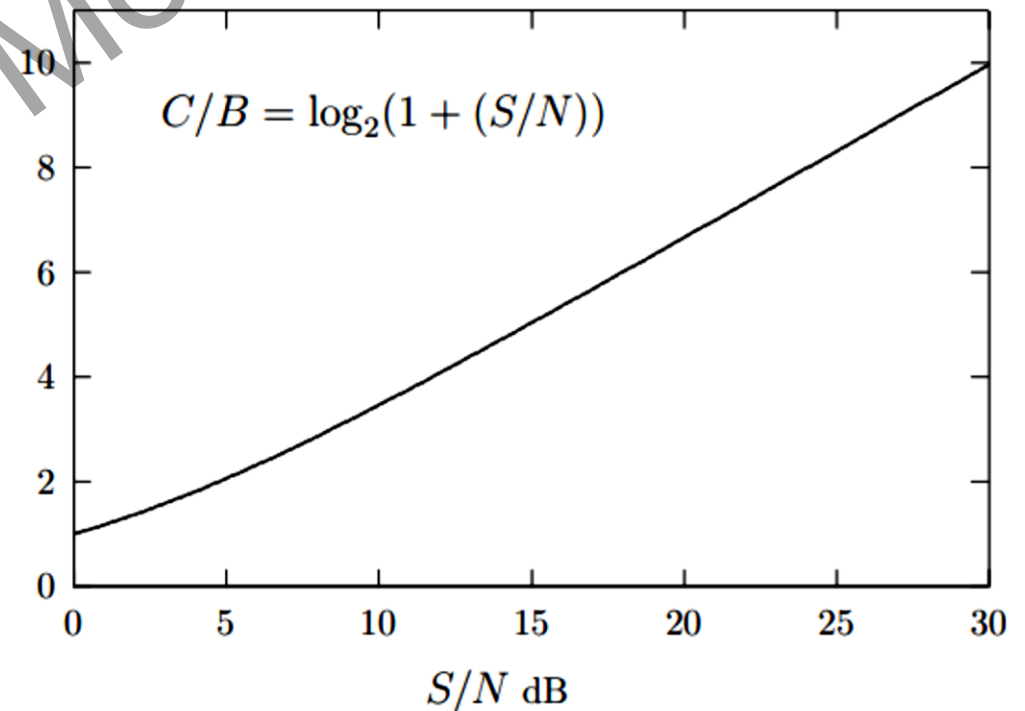
Communication System Parameters

Shannon's Equation (Channel Capacity)

- The **limitation imposed on communication** by the channel bandwidth and the SNR is given by,

$$C = B \log_2(1 + \text{SNR}), \text{ bit/s}$$

- ❖ The noise is AWGN (Additive White Gaussian Noise)
- C is the rate of information transmission per second, also known as channel capacity;
 - The maximum number of binary symbols (bits) that can be transmitted per second with a probability of error arbitrarily close to zero.
- Shannon's equation clearly brings out the limitation on the rate of communication imposed by B and SNR.



Communication System Parameters

Signal Power in Decibels (dB)

- The standard unit for **signal power** is watts (W).
- Often signal power is expressed on a **decibel scale**.

$$G_{\text{dB}} = 10 \log_{10} \frac{P_2}{P_1} \text{ dB}$$

- It is often useful to express power relative to a **reference power**.
 - For 1W as a reference:

$$P_{2,\text{dBW}} = 10 \log_{10} \frac{P_2}{1 \text{ W}} = 10 \log_{10} P_2 \text{ dBW}$$

- For 1mW as a reference:

$$P_{2,\text{dBm}} = 10 \log_{10} \frac{P_2}{1 \text{ mW}} = 10 \log_{10} P_2 \text{ dBm}$$

Communication System Parameters

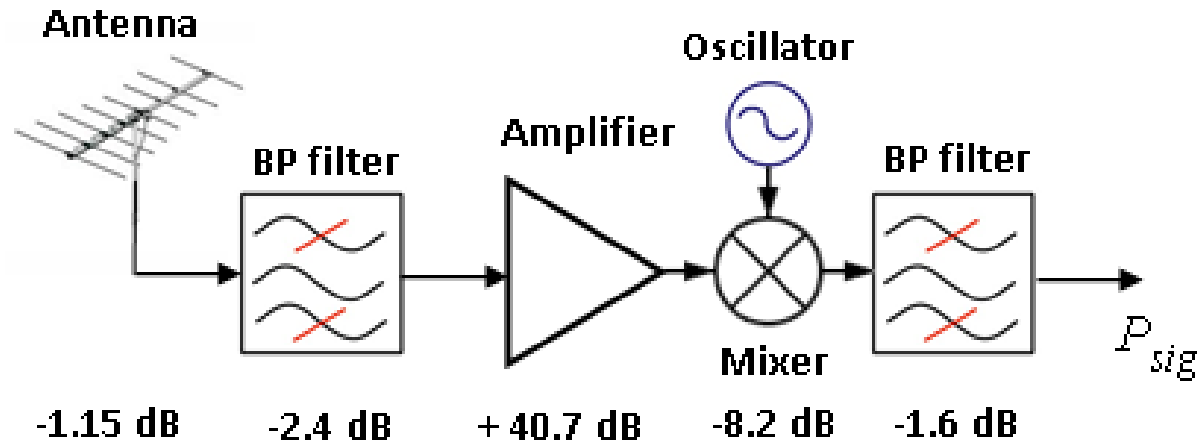
Example: Expressing Power Gain in dB

Suppose we have an amplifier that delivers a tenth of a watt (0.1 W) into the load resistor R_L when driven from a source delivering 2 milli-watts (0.002 W) at the amplifier's input, what is the gain in decibels?



$$G_{dB} = 10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \left(\frac{0.1 \text{ W}}{0.002 \text{ W}} \right)$$
$$G_{dB} = 10 \log_{10}(50) = 16.88 \text{ dB}$$

Note: 0.1 W in dBm is $10 \log_{10} \left(\frac{100 \text{ mW}}{1 \text{ mW}} \right) = 20 \text{ dBm}$



You are given the receiver block as schematically shown below. It consists of a receiving antenna which delivers a signal to a bandpass filter. The antenna feed has a loss of -1.15 dB and the filter's loss is -2.4 dB . The filter's output drives an amplifier with gain of $+40.7\text{ dB}$ and this RF signal is input to a mixer which performs a frequency translation to an IF signal (but the mixer has a loss of -8.2 dB). Finally, the IF signal travels through another bandpass filter with a nominal loss of -1.6 dB . If the signal strength output by the last filter is $P_{sig} = -86.3\text{ dBm}$, answer the following:

1. What is the signal strength of the signal P_{ant} at the antenna?
2. How many milliwatts is this signal power?

Communication Link: Gain & Loss

Communication System Parameters

Channel Capacity

- **Q1:** A telephone landline normally has a bandwidth of 3100 Hz (300 Hz to 3400 Hz) assigned for data communication. The signal-to-noise ratio (SNR) is usually of the order of 3100 numerically. From Shannon's limit what is the capacity in bits per second?
- **Q2:** A communication channel has a channel capacity of 24 Mbps with an electrical bandwidth of 2 MHz. What is the signal-to-noise ratio (SNR) in decibels?